1. Let P (x) denote the statement "x >7." Which of these have truth value true?
   a) P (0)
   b) P (4)
   c) P (6)
   d) P (9)

2. The statement," Every comedian is funny" where C(x) is "x is a comedian" and F (x) is "x is funny" and the domain consists of all people.
   a) $\exists x(C(x) \wedge F(x))$
   b) $\forall x(C(x) \wedge F(x))$
   c) $\exists x(C(x) \rightarrow F(x))$
   d) $\forall x(C(x) \rightarrow F(x))$

3. The statement, "At least one of your friends is perfect". Let P (x) be "x is perfect" and let F (x) be "x is your friend" and let the domain be all people.
   a) $\forall x\ (F(x) \rightarrow P(x))$
   b) $\forall x\ (F(x) \wedge P(x))$
   c) $\exists x\ (F(x) \wedge P(x))$
   d) $\exists x\ (F(x) \rightarrow P(x))$

4. Let A and B be two non-empty relations on a set S. Which of the following statements is false?
   a) A and B are transitive $\Rightarrow A \cap B$ is transitive
   b) A and B are symmetric $\Rightarrow A \cup B$ is symmetric
   c) A and B are transitive $\Rightarrow A \cup B$ is not transitive
   d) A and B are reflexive $\Rightarrow A \cap B$ is reflexive

5. Determine the characteristics of the relation aRb if $a^2 = b^2$.
   a) Transitive and symmetric
   b) Reflexive and asymmetry
   c) Trichotomy, antisymmetry, and irreflexive
   d) Symmetric, Reflexive, and transitive

6. How many binary relations are there on a set S with 9 distinct elements?
   a) 2^{90}
   b) 2^{100}
   c) 2^{81}
   d) 2^{60}

7. _____ number of reflexive relations are there on a set of 11 distinct elements.
   a) 2^{110}
   b) 3^{121}
   c) 2^{90}
   d) 2^{132}

8. The number of reflexive as well as symmetric relations on a set with 14 distinct elements is _____
   a) 4^{120}
   b) 2^{70}
   c) 3^{201}
   d) 2^{91}

9. The number of symmetric relations on a set with 15 distinct elements is _____
   a) 2^{196}
   b) 2^{50}
   c) 2^{320}
   d) 2^{78}

10. Suppose S is a finite set with 7 elements. How many elements are there in the largest equivalence relation on S?
    a) 56
    b) 78
    c) 49
    d) 100

11. How many elements are there in the smallest equivalence relation on a set with 8 elements?
    a) 10^2
    b) 8
    c) 48

d) 32

12. If a set A has 8 elements and a set B has 10 elements, how many relations are there from A to B?
a) 2^{90}
b) 3^{80}
c) 164
d) 2^{80}

13. Let a set S = {2, 4, 8, 16, 32} and <= be the partial order defined by S <= R if a divides b. Number of edges in the Hasse diagram of is _____
a) 6
b) 5
c) 9
d) 4

14. Suppose X = {a, b, c, d} and π_1 is the partition of X, π_1 = {{a, b, c}, d}. The number of ordered pairs of the equivalence relations induced by _____
a) 15
b) 10
c) 34
d) 5

15. R is a binary relation on a set S and R is reflexive if and only if _____
a) r(R) = R
b) s(R) = R
c) t(R) = R
d) f(R) = R

16. . If R_1 and R_2 are binary relations from set A to set B, then the equality _____ holds.
a) $(R^c)^c = R^c$
b) $(A \times B)^c = \Phi$
c) $(R_1 \cup R_2)^c = R_1^c \cup R_2^c$
d) $(R_1 \cup R_2)^c = R_1^c \cap R_2^c$

17. A relation R is defined on the set of integers as aRb if and only if a+b is even and R is termed as _____
a) an equivalence relation with one equivalence class
b) an equivalence relation with two quivalence classes
c) an equivalence relation
d) an equivalence relation with three equivalence classes

18. . The binary relation U = Φ (empty set) on a set A = {11, 23, 35} is _____
a) Neither reflexive nor symmetric
b) Symmetric and reflexive
c) Transitive and reflexive
d) Transitive and symmetric

19. . If $n(A \times B) = 6$ and $A = \{1, 3\}$ then $n(B)$ is
(1) 1
(2) 2
(3) 3
(4) 6

20. What does the Pigeonhole Principle state?
A) If you have more pigeons than pigeonholes, at least one pigeonhole must contain more than one pigeon.
B) If you have more pigeonholes than pigeons, each pigeonhole must contain at least one pigeon.
C) Every pigeon must have a unique pigeonhole.
D) Pigeons always occupy empty pigeonholes.

21. Suppose you have 13 pairs of socks in a drawer. What is the minimum number of socks you must take out (without looking) to ensure you get at least one matching pair?
a) 2
b) 3
c) 14
d) 27

22. If 13 people choose a birthday at random, what is the minimum number of people who must share the same birth month?

a) 0
b) 1
c) 2
d) 3

23. You have 6 pairs of shoes. What is the minimum number of shoes you must pick to guarantee at least one complete pair?
A) 6
B) 7
C) 11
D) 12

24. In a room of 367 people, at least two people must share the same.
A) Birthday
B) Favorite color
C) First name
D) Hobby

25. In a group of 100 people, each shaking hands with at least one other person, what is the minimum number of people who must have the same number of handshakes?
A) 1
B) 2
C) 50
D) 101

26. In a basket, there are 10 apples, 15 oranges, and 20 bananas. What is the minimum number of fruits you need to take to ensure at least two of the same kind?
A) 2
B) 3
C) 4
D) 5

27. A drawer contains socks of 10 different colors. What is the minimum number of socks you must pull out (at random) to be sure of having at least two socks of the same color?
a) 10
b) 11
c) 12
d) 20

28. Consider a set of 100 integers. What is the minimum number of integers you need to choose to guarantee that at least two of them have the same remainder when divided by 10?
a) 10
b) 11
c) 12
d) 100

29. . If ${}^{16}P_{r-1} : {}^{15}P_{r-1} = 16 : 7$ then find r.
a) 10
b) 12
c) 7
d) 8

30. . In a colony, there are 55 members. Every member posts a greeting card to all the members. How many greeting cards were posted by them?
a) 990
b) 890
c) 2970
d) 1980

31. Find the number of ways of arranging the letters of the words DANGER, so that no vowel occupies odd place.
a) 36
b) 48
c) 144
d) 96

32. . Find the sum of all four-digit numbers that can be formed by the digits 1, 3, 5, 7, 9 without repetition.
a) 666700
b) 666600
c) 678860
d) 665500

33. If ${}^{n}P_{r} = 3024$ and ${}^{n}C_{r} = 126$ then find n and r.
a) 9, 4
b) 10, 3
c) 12, 4
d) 11, 4

34. Find the number of rectangles and squares in an 8 by 8 chess board respectively.
a) 296, 204
b) 1092, 204

c) 204, 1092
d) 204, 1296

35. There are 20 points in a plane, how many triangles can be formed by these points if 5 are colinear?
a) 1130
b) 550
c) 1129
d) 1140

36. In how many ways can we select 6 people out of 10, of which a particular person is not included?
a) ${}^{10}C_3$
b) ${}^{9}C_5$
c) ${}^{9}C_6$
d) ${}^{9}C_4$

37. . ${}^{n}C_r$ = _____
a) n!
b) $n!r!$
c) $n!(n-r)!$
d) $n!(n-r)!r!$

38. ${}^{n}P_r = {}^{n}C_r$ * _____
a) r!
b) 1/r!
c) n!
d) 1/n!

39. If ${}^{n}C_2 = {}^{n}C_3$ then find n.
a) 2
b) 3
c) 5
d) 6

40. . If ${}^{6}C_2 = {}^{6}C_X$ then find possible values of x.
a) 2
b) 4
c) 2 and 4
d) 3

41. Determine n if ${}^{2n}C_3 : {}^{n}C_3 = 9:1$.
a) 7
b) 14
c) 28
d) 32

42. Out of a group of 5 persons, find the number of ways of selecting 3 persons.
a) 1
b) 5
c) 10
d) 15

43. Determine which of these conditional statements is false.
(a) If 1 + 1 = 2 then 2 + 2 = 5
(b) If 1 + 1 = 3 then 2 + 2 = 4
(c) If 1 + 1 = 3 then 2 + 2 = 5
(d) If monkeys can fly, then 1 + 1 = 3

44. The negation of the statement "Ibrahim is smart and hard working" is
(a) Ibrahim is not smart and hardworking.
(b) Ibrahim is not smart and not hardworking.
(c) Ibrahim is smart and not hardworking.
(d) Ibrahim is not smart or Ibrahim is not hardworking.

45. Let P: I am in Patiala.; Q: I love cricket.; then Q→ P is ?
a) If I love cricket then I am in Patiala
b) If I am in Patiala then I love cricket
c) I am not in Patiala
d) I love cricket

46. Let P: This is a great website, Q: You should not come back here. Then 'This is a great website and you should come back here.' is best represented by?
a) ~P V ~Q
b) P ∧ ~Q
c) P V Q
d) P ∧ Q

47. Let P: We should be honest., Q: We should be dedicated., R: We should be

overconfident. Then 'We should be honest or dedicated but not overconfident.' is best represented by?

a) ~P V ~Q V R

b) P ∧ ~Q ∧ R

c) P V Q ∧ R

d) P V Q ∧ ~R

Which of the following propositions is tautology?

A. (p v q) →q

B. p v (q→p)

C. p v (p→q)

D. Both (b) & (c)

48. Which of the following is a tautology

a) ~(p∧~p)

b) (p ∧ p)

c) (p∧~p)

d) None of these

49. Which of the following is not a proposition

a) Paris is in France.

b) 1+1=2

c) where are you going?

d) 9<6

50. Which of the following statements is the negation of the statements "4 is odd or -9 is positive"?
a) 4 is even or -9 is not negative
b) 4 is odd or -9 is not negative
c) 4 is even and -9 is negative
d) 4 is odd and -9 is not negative

51.) Which of the following represents: ~A (negation of A) if A stands for "I like badminton but hate maths"?
a) I hate badminton and maths
b) I do not like badminton or maths
c) I dislike badminton but love maths
d) I hate badminton or like maths

52. The contrapositive of p → q is the proposition:
a) ¬p → ¬q
b) ¬q → ¬p
c) q → p
d) ¬q → p

53. The inverse of p → q is the proposition:
a) ¬p → ¬q
b) ¬q → ¬p
c) q → p
d) ¬q → p

54. What is the contrapositive of the conditional statement? "The home team misses whenever it is drizzling?"
a) If it is drizzling, then home team misses
b) If the home team misses, then it is drizzling
c) If it is not drizzling, then the home team does not misses
d) If the home team wins, then it is not drizzling

55. What are the inverse of the conditional statement "If you make your notes, it will be a convenient in exams."
a) "If you make notes, then it will be a convenient in exams."
b) "If you do not make notes, then it will not be a convenient in exams."
c) "If it will not be a convenient in exams, then you did not make your notes."
d) "If it will be a convenient in exams, then you make your notes

56. The compound propositions p and q are called logically equivalent if _____ is a tautology.
a) p ↔ q
b) p → q
c) ¬ (p ∨ q)
d) ¬p ∨ ¬q

57. p ∨ q is logically equivalent to:
a) ¬q → ¬p
b) q → p
c) ¬p → ¬q
d) ¬p → q

58. Which of the following statement is correct?
a) p ∨ q ≡ q ∨ p
b) ¬(p ∧ q) ≡ ¬p ∨ ¬q
c) (p ∨ q) ∨ r ≡ p ∨ (q ∨ r)
d) All of mentioned

59. If the proposition 'No men are honest' is taken to be false which of the following

proposition/propositions can be claimed certainly to be true?
Propositions:
(A) All men are honest
(B) Some men are honest
(C) Some men are not honest
(D) No honest person is man

60. If the statement 'None but the brave wins the race' is false which of the following statements can be claimed to be true?
Select the correct code:
(A) All brave persons win the race.
(B) Some persons who win the race are not brave.
(C) Some persons who win the race are brave.
(D) No person who wins the race is brave.

61. $p \wedge q$ is True when
a) p is true, q is false
b) p is false, q is true
c) p is true, q is true
d) p is false, q is false

62. De Morgan Law states that
a) $\sim(p \wedge q) = \sim p \vee \sim q$
b) $\sim(p \wedge q) = \sim p \wedge \sim q$
c) $(p \wedge q) = \sim p \vee \sim q$
d) None of these

63. Which of the following statements is the negation of the statements "4 is odd or -9 is positive"?
a) 4 is even or -9 is not negative
b) 4 is odd or -9 is not negative
c) 4 is even and -9 is negative
d) 4 is odd and -9 is not negative

64. How many 2 digit numbers can be formed from the digits 1, 2, 3, 4, 5, 6 if the digits can be repeated?
(a) 25 (b) 24
(c) 36 (d) 12

65. How many 2 digit numbers can be formed from the digits 1, 2, 3, 4 and 5, if the repetition of digits is not allowed?
(a) 25 (b) 20
(c) 60 (d) 12

66. How many 2 digit odd numbers can be formed from the digits 1, 2, 3, 4 and 5, if the repetition of digits is allowed?
(a) 25 (b) 20
(c) 15 (d) 10

67. The final value of x where $x = 10! - 9!$ is
(a) $10! \times 9$ (b) $9! \times 10$
(c) $9! \times 9$ (d) none of these

68. The final value of x where $x = \frac{8!}{6! \times 2!}$ is
(a) 36 (b) 26
(c) 28 (d) 12

69. The value of $\frac{n!}{(n-r)!}$ is, when $n = 9, r = 5$
(a) 15130 (b) 26510
(c) 28120 (d) 15120

70. How many number of 4 letter words, with or without meaning, can be formed out of the letters of the word 'WATCH' ?
25 (b) 240 (c) 160 (d) 120

71. The number of permutations of the letters of the word 'FILL' is
(a) 8 (b) 12
(c) 16 (d) 24

72. In how many different ways can the letters of the word 'LEADING' be arranged in such a way that the vowels always come together?

(a) 720 (b) 360 (c) 480 (d) 5040

73. If the repetition of the digits are allowed, then the no. of even natural numbers having three digit is

(a) 250 (b) 350
c) 450 (d) 550

74. The number of arrangement of the letter of the word BANANA in which the two N's do not appear adjacently is

(a) 40 (b) 60 (c) 80 (d) 100

75. 6 men and 4 women are to be seated in a row such that no two women seat together. The no. of ways they can be seated is

a) 604800 b) 17280 c) 120960 d) 518400

76.If the letter of the word SACHIN are arranged in all possible ways and these words are written in as dictionary, then the word SACHIN appear at serial no 601 b) 600 c) 602 d) 603

77. The number of chords that can be drawn through 21 points on a circle

a) 210 b) 21! c) 420 d) None of these

78. How many 2 digit even numbers can be formed from the digits 1, 2, 3, 4, 5 if the digits are repeated

(a) 25 b) 20 c) 10 d) None of these

79. How many 4 letter words, with or without meaning, can be formed out of the letters of the word 'LIke' without repeating letters

(a) 256 b) 24 c) 16 d) 32

80. How many 4 letter words, with or without meaning, can be formed out of the letters of the word 'MATCH' without repeating any letter?

(a) 25 b) 240 c) 160 d) 120

81. In How many ways can the letters of the word LEADER be arranged

(a) 120 (b) 360 (c) 720 (d) 1440

82. A polygon has 44 diagonal then the no of its sides ar (a) 13 (b) 15 (c) 11 (d) 9

83. The no triangles that can be formed from 12 points 7 of which are collinear ?

(a) 131 (b) 185 (c) 161 (d) 241

84. In how many ways can 10 persons be arranged in a circle ?

(a) 9! (b) 10! (c) 11! (d) 19!

85. In how many ways can n prizes be given to n students so each is to receive one prize ?

(a) $n!$ (b) (n-1)! (c) 2n! (d) $(n+1)!$

86. In how many ways can n prizes be given to n students so each may receive any no. of prizes ?

(a) $n!$ (b) (n-1)! (c) $n^n!$ (d) n^n

87.In how many ways can n prizes be given to n students so no one can receive all the prizes ?

(a) $n!$ (b) (n-1)! (c) $n^n - 1$ (d) $n^n - n$

88.In how many ways can a team of 11 players be selected out of 15 players so that one particular player is always be there?

(a) 1001 (b) 1365 (c) 2535 (d) 2550

89. In how many ways can a team of 11 players be selected out of 15 players excluding one particular player ?

(a) 1001 (b) 1365 (c) 364 (d) 550

90 The relation { (1,2), (1,3), (3,1), (1,1), (3,3), (3,2), (1,4), (4,2), (3,4)} is

a) Reflexive

b) Transitive

c) Symmetric

d) Asymmetric

91 If every two elements of a poset are comparable then the poset is called

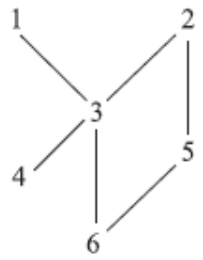
a) sub ordered poset

b) totally ordered poset

c) sub lattice

d) semigroup

92 The maximal and minimal elements of Poset are



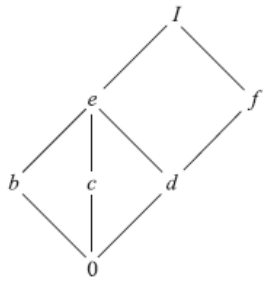
a) Maximal 5, 6; Minimal 1

b) Maximal 4, 6; Minimal 1

c) Maximal 1, 2; Minimal 4, 6

d) Maximal 4, 6; Minimal 2

93 The complement of e and f



a) None and b,c

b) f and e

c) None and d

d) d and b,c

94 Let a set S = {2, 4, 8, 16, 32} and <= be the partial order defined by S <= R if a divides b. Number of edges in the Hasse diagram of is _____

a) 6

b) 5

c) 9

d) 4

95 A _____ has a greatest element and a least element which satisfy 0<=a<=1 for every a in the lattice (say, L).

a) semilattice

b) join semilattice

c) meet semilattice

d) bounded lattice

96 Consider binary a relation R={(a,b)|a=b} on the set of integers. Then which of the following options is true?

a) R is reflexive only
b) R is symmetric only
c) R is reflexive and anti-symmetric
d) R is reflexive, symmetric, anti-symmetric, and transitive

97 R={(a,b)|a=b+1}. Which of the following options is true?

a) R is reflexive

b) R is anti-symmetric

c) R is a partial order relation

d) R is equivalence relation

98 Find the greatest lower bound and the least upper bound of the sets {1, 2, 3, 4, 5}, if they exist, in the poset (Z+, |)

A) 1, does not exit

B) 1, 5

c) does not exist, 5

d) does not exit, does not exit

99 A lattice is a poset in which

a) every pair has LUB and GLB
b) every element has complement
c) every element has inverse
d) none of these

100. A lattice with greatest element 1 and least element 0 is called

a) distributive lattice
b) bounded lattice
c) modular lattice
d) complete lattice

101 If every pair of elements in a poset is comparable, it is called

a) lattice

b) chain

c) equivalence relation

d) semigroup

102 Total number of symmetric relations on a set of 7 distinct elements are

a) 2^{30}

b) 2^{36}

c) 2^{28}

d) 2^{35}

103 Let A={1,2,3}
LetR={(1,1),(2,2),(3,3),(1,2),(2,1)}
The relation R is:
a) Reflexive and symmetric
b) Reflexive and antisymmetric
c) Symmetric and transitive
d) Transitive only

104 Which of the following relations on Z is **not transitive**?

a) $a \leq b$
b) $a=b$
c) $a<b$
d) $a=b+1$

105 In the poset (Z+,|) the numbers **6 and 15** are

a) Comparable
b) Not comparable
c) Equal
d) Comparable but not ordered

106 In Boolean algebra

$a+a=a$

This law is called

a) Identity law
b) Idempotent law
c) Complement law
d) Absorption law

107 Which of the following is true in Boolean algebra?

a) $a+a'=0$
b) $a+a'=1$
c) $aa'=1$
d) $a+1=0$

108 A lattice in which **every element has a complement** is called

a) Distributive lattice
b) Complemented lattice
c) Modular lattice
d) Bounded lattice

109 In a complemented lattice, if b is a complement of a, then

a) $a \vee b=a$
b) $a \wedge b=a$
c) $a \vee b=1$ and $a \wedge b=0$
d) $a=b$

110 The dual of $a+0=a$ is

a) $a \cdot 0=a$
b) $a \cdot 1=a$
c) $a+1=a$
d) $a \cdot a=a$

111 If **10 pairs of socks** are in a drawer, what is the **minimum number of socks** that must be selected to guarantee at least one matching pair?

a) 2
b) 10
c) 11
d) 12